

Brightfield-only single-nucleus differentiation prediction with deep-ensemble uncertainty

Technical Report — AI4S Open Innovation

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Abstract

Label-free prediction of cellular state can reduce dependence on destructive or endpoint fluorescence measurements in high-throughput 3D cell culture experiments. Using the public Bright2Nuc dataset, we develop a brightfield-only pipeline for predicting a per-nucleus differentiation score under culture-wise out-of-fold evaluation. We compare a feature-based random forest (M0), a single-task ResNet18 (M1), a multitask ResNet18 (M2), a multitask ConvNeXt-Tiny (M3), and a frozen DINOv2 representation followed by ridge regression (M4). The strongest individual model, M3, reaches nucleus-level $R^2 = 0.693$ and $MAE = 0.121$. A deep ensemble of five independently trained M3 models per fold further improves nucleus-level performance to $R^2 = 0.718$ and $MAE = 0.117$, while providing an uncertainty estimate through ensemble disagreement. Uncertainty is positively associated with absolute error (Spearman $\rho = 0.519$); rejecting the 20% most uncertain nuclei yields $R^2 = 0.827$ and $MAE = 0.084$ on the retained set. A separate longitudinal sensitivity analysis on six cultures indicates representation-dependent temporal trajectories and associations with acquisition characteristics. Without matched longitudinal fluorescence ground truth, these trajectories are interpreted as sensitivity analyses rather than direct measurements of biological dynamics.

1 Problem and motivation

Microfluidic 3D stem-cell culture platforms enable large-scale experiments, while repeated fluorescence-based phenotyping is difficult to apply to the same living cultures over time. The Bright2Nuc study introduced a label-free deep-learning framework for extracting nuclear information from confocal brightfield images and demonstrated downstream segmentation, cell-state prediction, and tracking in 3D human induced pluripotent stem-cell (hiPSC) cultures differentiating toward definitive endoderm [1].

The published feature-based differentiation-state benchmark reported $R^2 = 0.48$ and $MAE = 0.19$ on 26,213 nuclei from 49 cultures [1]. Our objective is to improve predictive accuracy while building a brightfield-only per-nucleus prediction system with held-out-culture validation and an informative uncertainty estimate. Fluorescence-derived quantities provide supervision during training but are not required at prediction time.

Our frozen static comparison comprises 30,877 nuclei from 48 cultures. Unlike the original published benchmark, which split individual nuclei randomly across five folds, our evaluation holds out entire cultures. The published numbers therefore provide biological and methodological context, not a strictly like-for-like performance comparison.

2 Data and target definition

We first describe the source data and the fluorescence-derived training target, before detailing the label-free brightfield input used by the predictive models.

2.1 Source data and experimental support

The project uses the public Bright2Nuc image data from the [Zenodo dataset](#), together with the original study and its open-source analysis resources [1]. The experiments comprise on-chip 3D hiPSC cultures imaged by confocal brightfield microscopy, with matched fluorescence measurements available for fixed cultures. The supervised benchmark used here comprises 30,877 nuclei from 48 cultures after applying the frozen local data-selection and target-construction pipeline.

The publication’s reported feature-based benchmark involved a different set of 26,213 nuclei from 49 cultures. Its original 120-feature representation contained 56 brightfield texture features and 64 morphological features derived from nuclear segmentation [1]. The local M0 experiment is evaluated on the separate frozen 48-culture support and is not presented as an exact reproduction of the published feature experiment.

2.2 Differentiation-label definition

The supervised target is a continuous differentiation label (DL) defined in the original study from three fluorescence-derived transcription-factor expression measurements: the pluripotency marker OCT4 and the differentiation markers FOXA2 and SOX17. With e_j denoting each normalized expression measurement, the published definition is [1]

$$\text{DL} = \frac{1}{1 + \frac{2e_{\text{OCT4}}}{e_{\text{FOXA2}} + e_{\text{SOX17}}}}. \quad (1)$$

In the original study, marker intensities were normalized using the first and ninety-ninth percentiles. The label increases as differentiation-marker expression dominates over OCT4 expression. It is a marker-derived proxy for differentiation state, not a direct measurement of all aspects of cellular identity or a time variable. The frozen local target table is shared across the static model comparisons; a target-consistency check between the original M0–M3 evaluation and M4 found a maximum absolute discrepancy below 6×10^{-8} .

OCT4, FOXA2, and SOX17 measurements also serve as auxiliary training targets for the multitask models. Neither these marker measurements nor the differentiation label are supplied as model inputs at inference time. All reported supervised accuracy is measured against the fluorescence-derived target on held-out cultures.

2.3 Brightfield representation and scope

Each nucleus is represented by a local 3D brightfield stack with shape $16 \times 64 \times 64$ (depth, height, width). The crop preserves local image context around a nucleus while limiting the input size. The learned regression models operate on these per-nucleus crops; their static metrics measure crop-to-score prediction rather than a separate end-to-end evaluation of upstream nucleus localization and segmentation. This distinction matters when considering deployment on previously unseen full microscope volumes.

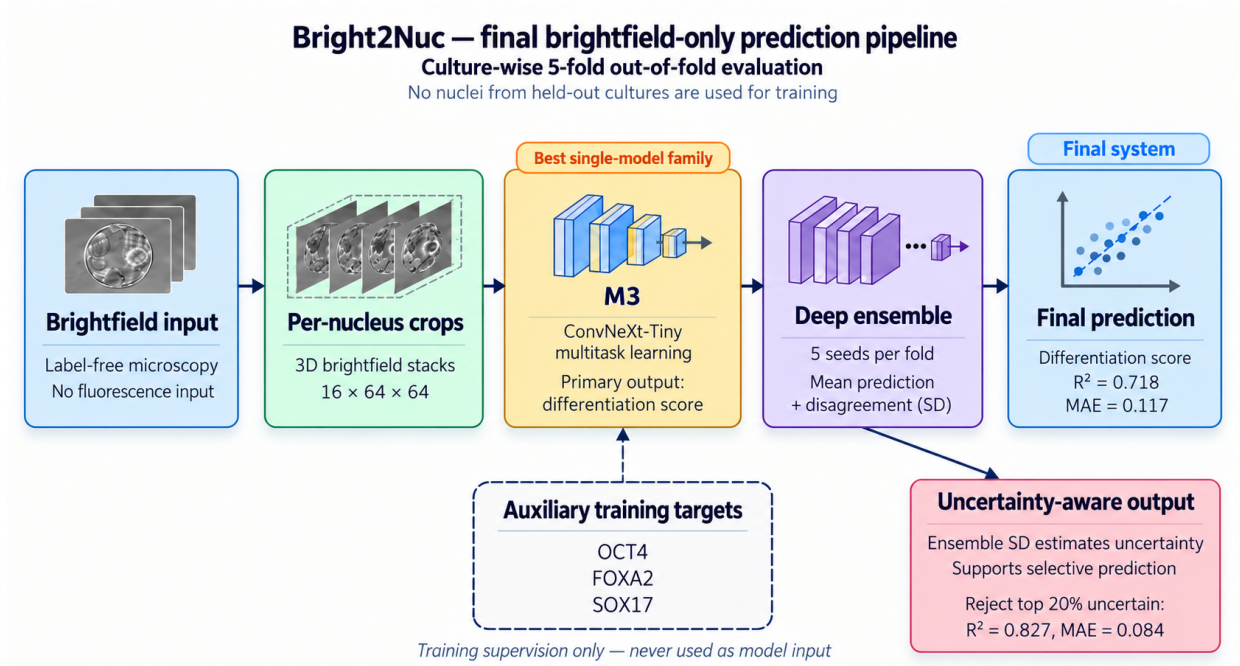


Figure 1: Final brightfield-only prediction pipeline. Brightfield crops are processed by M3, a multitask ConvNeXt-Tiny, and a five-member deep ensemble in each held-out fold. OCT4, FOXA2, and SOX17 are auxiliary training targets only. The inference outputs are the predicted differentiation score and an ensemble-disagreement uncertainty estimate.

3 Evaluation protocol

We describe how held-out-culture generalization is assessed, how out-of-fold predictions are assembled, and how model performance and paired improvements are quantified.

3.1 Culture-wise cross-validation

All principal static results use the same frozen five-fold culture-wise out-of-fold (OOF) evaluation. Cultures, rather than individual nuclei, are assigned to folds. For each outer fold, model fitting, preprocessing-statistic estimation, and any internal model-selection operations use only the corresponding development cultures; nuclei from the held-out cultures are reserved for evaluation. For M1–M3, the development cultures are further divided into training and validation sets at the culture level. Stratification uses a categorical label derived from the culture-level mean differentiation score, as implemented in the frozen split-generation code, so that the continuous score itself is not treated as a class label. The internal train/validation split remains fixed across the M3 deep-ensemble members in a given outer fold.

Each eligible nucleus is predicted by a model trained without its culture. Concatenating the five held-out predictions yields the global OOF evaluation set of 30,877 nuclei from 48 cultures. This grouped design limits leakage through culture-specific acquisition and morphological signatures. It estimates generalization to other cultures from the represented data source, not necessarily to new microscopes, laboratories, or cell lines.

Unless explicitly stated otherwise, the training hyperparameters reported in this work correspond to fixed development configurations used for the final experiments; no exhaustive hyperparameter search was performed.

3.2 Metrics and aggregation

For reference targets y_i , predictions $\hat{y}_i$, and N test nuclei, the main pointwise metrics are

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (2)$$

Pearson correlation measures linear association; Spearman correlation measures rank association. All global nucleus-level metrics are calculated after concatenating the OOF predictions, rather than by averaging fold-level metric values.

At the culture level, reference targets and predictions are each averaged within a culture. The same four metrics are then calculated across the resulting 48 culture-level pairs. These two evaluation levels answer distinct questions: nucleus-level metrics assess individual predictions, whereas culture-level metrics assess differences among culture means. Fold-wise means and standard deviations are provided separately in Appendix A and should not be confused with the pooled global OOF values.

3.3 Paired model comparison and bootstrap analysis

M3 and M2 are compared on matching held-out cultures. The frozen paired analysis shows lower error for M3 in 45 of 48 cultures and for M2 in three cultures. A paired Wilcoxon signed-rank test is used for the culture-wise comparison. To quantify uncertainty in the global performance difference while respecting within-culture dependence, the bootstrap resamples cultures, carrying all nuclei from each sampled culture together, and reports 95% intervals for the M3–M2 differences in pooled nucleus-level R^2 and MAE. Individual nuclei are therefore not treated as independent bootstrap units. The bootstrap implementation and random seed are fixed in the released analysis code.

3.4 Deep-ensemble uncertainty analysis

For the deep ensemble, uncertainty is evaluated on the same 30,877 OOF nuclei used for accuracy assessment. For each nucleus, the ensemble mean is the final prediction and the sample standard deviation across the five member predictions is the disagreement score. Its association with absolute error is quantified using Pearson and Spearman correlations. Predictions are additionally divided into ten uncertainty deciles, and selective prediction is evaluated by rejecting nuclei in descending order of ensemble disagreement and recomputing metrics only on the retained subset. These analyses evaluate ranking and triage utility; they do not constitute formal probabilistic calibration.

4 Model families

Five model families span pre-extracted features, end-to-end crop-based regression, multitask learning, and frozen pretrained visual representations. This comparison distinguishes the contribution of learned representations from that of auxiliary supervision and backbone choice.

4.1 M0: feature-based random forest

M0 is the non-deep-learning regression baseline [2]. The final M0 model operates on 119 pre-extracted features released with the Bright2Nuc data: 63 morphology-derived features from `Features.csv` and 56 brightfield-derived features from `Features_Brightfield.csv`.

A second diagnostic variant additionally included the `boarder_distance` variable, yielding 120 features. The frozen 119-feature configuration is retained as the primary local M0 baseline, while

the 120-feature version is treated only as a diagnostic sensitivity variant; consequently, M0 is not presented as an exact reproduction of the published 120-feature benchmark.

For each outer culture-wise fold, a `RandomForestRegressor` is trained on all nuclei from the remaining cultures using 1000 trees, a maximum tree depth of 10, squared-error splitting, bootstrap sampling, `max_features=1.0`, and random seed 42. No feature standardization is applied.

The frozen five-fold partition is generated at the culture level using stratified grouped cross-validation. Cultures are stratified according to their mean differentiation label, and all nuclei from a given culture remain in the same fold.

M0 therefore provides a conventional feature-based reference against which the benefit of learned image representations can be assessed. It should not be confused with the published Bright2Nuc benchmark, which used a different 120-feature configuration, a different sample support, and nucleus-level random splitting.

4.2 M1: ResNet18 single-task model

M1 replaces the feature-based baseline with an end-to-end ResNet18 [3] trained to predict the scalar differentiation label directly from local brightfield stacks.

An ImageNet-pretrained ResNet18 backbone is adapted to accept the 16 brightfield slices as input channels. Thus, the pretrained three-channel input stem is converted to a 16-channel stem during model construction, while the remaining pretrained encoder weights are retained. Each $16 \times 64 \times 64$ per-nucleus brightfield crop is spatially resized to 128×128 using bilinear interpolation before entering the encoder. Global-average-pooled features are passed to a single linear regression head.

Brightfield intensities are standardized using a global mean and standard deviation estimated exclusively from the training subset of the corresponding outer fold. No validation or test images contribute to these normalization statistics.

During training, spatial augmentation is restricted to the XY plane and comprises rotations by multiples of 90° and horizontal reflections. Additional intensity augmentation includes random contrast and brightness factors in $[0.8, 1.2]$, gamma transformation in $[0.8, 1.2]$, and additive Gaussian noise with standard deviation 0.01. The z-axis is not reversed.

The regression objective is a SmoothL1 loss. Optimization uses AdamW with learning rates of 10^{-4} for the pretrained encoder and 3×10^{-4} for the regression head, with weight decay 0.05. A microbatch size of 64 and gradient accumulation over four steps give an effective batch size of 256. Training uses mixed precision, gradient clipping at norm 1.0, one epoch of learning-rate warm-up followed by cosine decay, and a maximum of 30 epochs.

Within each outer development set, an internal culture-level validation subset comprising 15% of the development cultures is selected using stratification based on culture-level mean differentiation score. Early stopping uses validation MAE with a patience of five epochs. The held-out outer fold is never used for normalization, checkpoint selection, or early stopping.

The selected checkpoint is evaluated using eight-view test-time augmentation, formed by four in-plane rotations and their horizontally reflected counterparts. Their predictions are averaged to obtain the final M1 prediction.

4.3 M2: ResNet18 multitask model

M2 retains the same ImageNet-pretrained ResNet18 backbone adapted to 16 input channels, spatial resize, brightfield normalization, data augmentation, optimization schedule, internal culture-level validation split, early stopping, and eight-view test-time augmentation used by M1. The architectural difference is a second linear head that predicts three auxiliary fluorescence-derived targets: OCT4,

FOXA2, and SOX17.

For each outer fold, the auxiliary marker targets are normalized using the first and 99th percentiles estimated exclusively from the M2 training cultures; normalized values are clipped to $[0, 1]$. The differentiation target itself remains the primary supervised endpoint. Both the primary and auxiliary losses use SmoothL1 regression, with

$$\mathcal{L}_{M2} = \mathcal{L}_{DL} + 0.5 \mathcal{L}_{\text{markers}}. \quad (3)$$

The marker-loss weight is therefore fixed at 0.5. M2 uses AdamW with encoder and head learning rates of 10^{-4} and 3×10^{-4} , respectively, weight decay 0.05, a microbatch size of 64, gradient accumulation over four steps, and an effective batch size of 256. Training is limited to 30 epochs with patience five. Importantly, checkpoint selection is based only on validation MAE for the primary differentiation task; auxiliary-marker validation performance does not select the checkpoint.

The final held-out prediction is the mean of the same eight in-plane rotation/reflection views used for M1. Fluorescence-derived quantities are used only as targets during training and evaluation of the auxiliary heads; they are never supplied as model inputs.

4.4 M3: ConvNeXt-Tiny multitask model

M3 retains the multitask target structure of M2 but replaces the ResNet18 backbone with an ImageNet-pretrained ConvNeXt-Tiny [4]. Its pretrained input stem is adapted during model construction to accept the 16 brightfield slices as input channels. The model therefore accepts a per-nucleus $16 \times 64 \times 64$ brightfield input, spatially resized to 128×128 , and predicts the primary differentiation label alongside OCT4, FOXA2, and SOX17.

The primary and auxiliary objectives are SmoothL1 losses with the same auxiliary weighting as M2:

$$\mathcal{L}_{M3} = \mathcal{L}_{DL} + 0.5 \mathcal{L}_{\text{markers}}. \quad (4)$$

Training and preprocessing. Brightfield intensities are standardized using statistics computed exclusively on the training subset of each fold. Auxiliary marker targets are normalized using the training-set first and 99th percentiles, with values clipped to $[0, 1]$. M3 uses the same XY rotations/reflections and intensity augmentations as M1 and M2.

M3 is optimized using AdamW, with learning rates of 10^{-4} for the encoder and 3×10^{-4} for the prediction heads, and a weight decay of 0.05. A microbatch size of 32 and gradient accumulation over eight steps yield an effective batch size of 256. Training uses mixed precision, gradient clipping at norm 1.0, one epoch of warm-up followed by cosine decay, a maximum of 30 epochs, and early stopping after five epochs without improvement in validation differentiation MAE. The checkpoint is selected using the primary validation task without test-time augmentation.

Test-time augmentation. Held-out predictions are obtained using eight-view test-time augmentation, comprising four in-plane rotations and their horizontally flipped counterparts. The eight predictions are averaged to produce the final M3 output.

4.5 M4: frozen DINOv2 representation

M4 provides an independent pretrained-representation comparison [5]. The frozen encoder is the DINOv2 ViT-S/14 representation used through `timm` (model identifier `vit_small_patch14_dinov2.lvd142m`). Three central brightfield slices, with zero-based indices 7, 8, and 9, are used as the

three channels of a pseudo-RGB image. The resulting image is resized to 518×518 using bicubic interpolation and normalized according to the pretrained encoder’s expected input preprocessing.

The encoder is not fine-tuned on Bright2Nuc. Its 384-dimensional CLS embedding is extracted for each nucleus. Within each outer culture-wise fold, embedding standardization and the ridge-regression head are fitted using only the corresponding development data. The ridge regularization strength is selected by culture-wise internal validation from

$$\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000\}.$$

The selected ridge model is then evaluated on the held-out cultures. M4 therefore tests whether a frozen generic visual representation can support the differentiation task without end-to-end fine-tuning. It is an independent representation control, not a model assumed to be invariant to acquisition artifacts; contrast, focus, density, and other acquisition-related changes can still affect its embeddings.

4.6 Deep ensemble and inference outputs

The final ensemble follows the deep-ensemble principle [6]. Five M3 instances are trained independently for each outer fold using training seeds 42–46. The outer culture-wise fold assignment and the internal train/validation partition are held fixed across ensemble members; specifically, the training seed changes while the split seed remains fixed at 42. Consequently, member differences reflect training stochasticity rather than different data partitions.

Each member uses the same architecture, preprocessing, optimization settings, early-stopping criterion, and eight-view test-time augmentation as the frozen M3 reference. Five outer folds and five training seeds therefore yield 25 trained models in total.

For member predictions $\hat{y}_i^{(1)}, \dots, \hat{y}_i^{(5)}$, the reported prediction and uncertainty are

$$\bar{y}_i = \frac{1}{5} \sum_{s=1}^5 \hat{y}_i^{(s)}, \quad \nu_i = \sqrt{\frac{1}{4} \sum_{s=1}^5 \left(\hat{y}_i^{(s)} - \bar{y}_i \right)^2}. \quad (5)$$

The standard deviation is the sample standard deviation (`ddof=1`). The quantity ν_i measures disagreement among trained members; it is an empirical predictive-uncertainty signal, not a formally calibrated posterior uncertainty or a guaranteed error bound.

5 Static prediction results

We compare the static OOF performance of the five model families and then assess how deep ensembling changes nucleus-level and culture-level accuracy.

5.1 Model comparison

The feature-based M0 baseline reaches nucleus-level $R^2 = 0.249$ and $\text{MAE} = 0.234$. Learned image representations improve performance with M1 ($R^2 = 0.403$, $\text{MAE} = 0.220$); multitask supervision provides a further gain with M2 ($R^2 = 0.453$, $\text{MAE} = 0.207$). Replacing ResNet18 with ConvNeXt-Tiny gives the largest improvement among individual models: M3 reaches $R^2 = 0.693$ and $\text{MAE} = 0.121$. The frozen DINOv2 representation M4 reaches $R^2 = 0.450$ and $\text{MAE} = 0.210$, close to M2 but below M3 on the primary nucleus-level endpoint.

Table 1: Static out-of-fold performance. The deep ensemble is the final predictive system. Bold values mark the strongest result in each metric column.

Model	Nucleus level				Culture level			
	R^2	MAE	Pearson	Spearman	R^2	MAE	Pearson	Spearman
M0 Random Forest	0.249	0.234	0.507	0.432	0.337	0.244	0.667	0.666
M1 ResNet18	0.403	0.220	0.641	0.566	0.522	0.219	0.851	0.802
M2 ResNet18 multitask	0.453	0.207	0.681	0.633	0.552	0.211	0.846	0.832
M3 ConvNeXt multitask	0.693	0.121	0.836	0.765	0.895	0.085	0.954	0.834
M4 DINOv2 + Ridge	0.450	0.210	0.672	0.602	0.678	0.177	0.934	0.839
M3 Deep Ensemble	0.718	0.117	0.848	0.777	0.892	0.089	0.956	0.855

Across the 48 held-out cultures, the paired M3–M2 comparison favors M3 in 45 cultures and M2 in three. The paired Wilcoxon test yields $p = 8.09 \times 10^{-11}$. Culture-level resampling gives an M3–M2 difference in pooled nucleus-level R^2 of $+0.245$ (95% bootstrap CI $[+0.150, +0.327]$) and in MAE of -0.087 (95% bootstrap CI $[-0.110, -0.063]$). These paired analyses complement the rounded global scores and explicitly account for culture-level dependence.

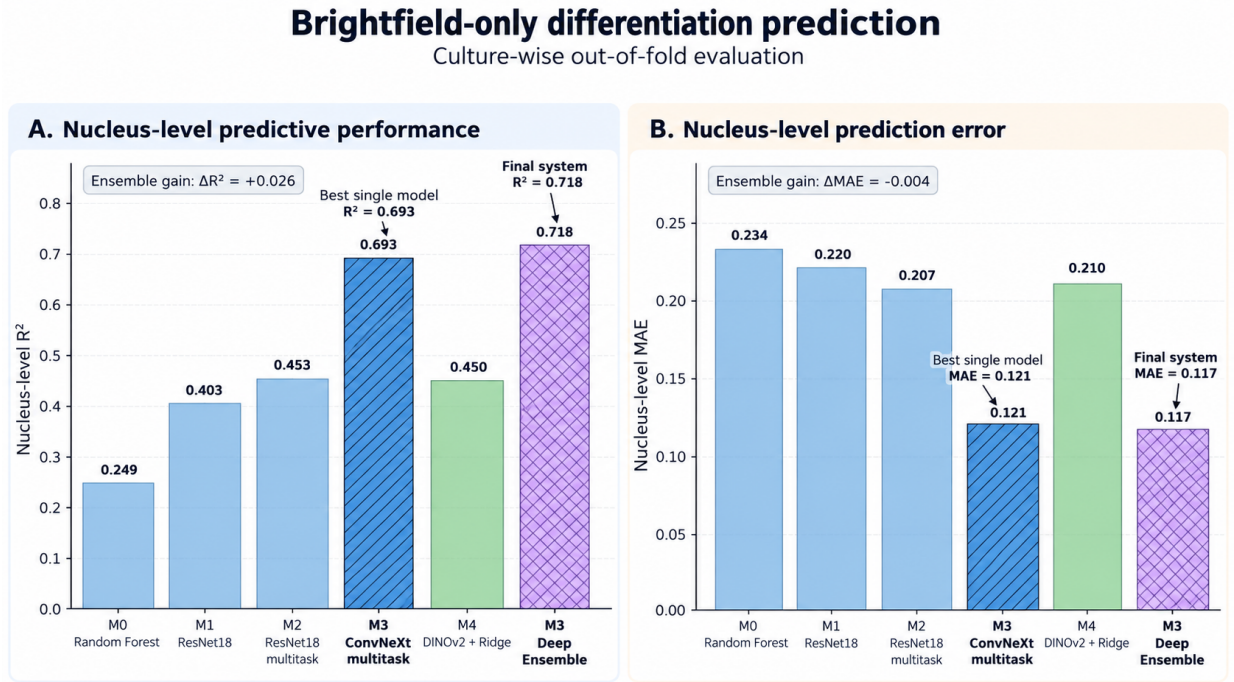


Figure 2: Brightfield-only differentiation prediction across model families. Nucleus-level OOF R^2 and MAE under culture-wise cross-validation for M0–M4 and the M3 deep ensemble. The ensemble improves R^2 from 0.693 to 0.718 and MAE from 0.121 to 0.117 relative to the original M3 single-model reference. No confidence intervals are displayed.

5.2 Effect of deep ensembling

Relative to M3 seed 42, the deep ensemble increases nucleus-level R^2 by $+0.0256$ and reduces MAE by 0.00424. At the culture level, the two predictions are similar: M3 reaches $R^2 = 0.895$ and MAE = 0.085, whereas the ensemble reaches $R^2 = 0.892$ and MAE = 0.089. The ensemble is therefore more accurate at the nucleus level without being uniformly superior under every aggregation metric.

For individual nuclei, the ensemble mean reaches Pearson = 0.848 and Spearman = 0.777. Its second benefit is the per-nucleus disagreement estimate used for error triage and selective prediction. Appendix B reports the five seed-specific OOF performances; the ensemble is retained rather than

selecting the best seed retrospectively on held-out predictions.

5.3 Auxiliary marker outputs

The M3 multitask head also predicts its three auxiliary targets. Its OOF marker-specific performance is $R^2 = 0.552$ and $\text{MAE} = 0.113$ for OCT4, $R^2 = 0.516$ and $\text{MAE} = 0.137$ for FOXA2, and $R^2 = 0.430$ and $\text{MAE} = 0.140$ for SOX17. These are auxiliary supervised outcomes, not fluorescence inputs at inference time, and do not substitute for validation of the primary differentiation-label regression.

6 Predictive uncertainty

We assess whether ensemble disagreement identifies more error-prone nuclei and whether uncertainty-based rejection improves accuracy on the predictions that remain available.

6.1 Uncertainty–error association

Across all 30,877 OOF nuclei, ensemble standard deviation is positively associated with absolute error of the ensemble-mean prediction (Spearman $\rho = 0.519$; Pearson $r = 0.493$). The scatter plot displays a random subset of 12,000 nuclei for readability; both reported correlations are calculated on all 30,877 OOF predictions. The correlation p-values underflow numerically at this sample size and are therefore not reported as $p = 0$.

When predictions are grouped into ten increasing uncertainty deciles, mean absolute error rises monotonically from 0.046 in the least uncertain decile to 0.285 in the most uncertain decile. The complete decile values are provided in Appendix C. This monotonic relationship indicates that ensemble disagreement is useful for ranking predictions by observed error risk, without implying probabilistic calibration.

6.2 Selective prediction

Table 2: Selective prediction on retained OOF nuclei after rejecting the largest ensemble standard deviations.

Rejected (%)	Retained nuclei	MAE	R^2
0	30,877	0.1168	0.7181
5	29,333	0.1069	0.7489
10	27,789	0.0981	0.7785
20	24,701	0.0844	0.8269

Rejecting the 20% most uncertain predictions reduces MAE from 0.117 to 0.084 and increases R^2 from 0.718 to 0.827 among the 24,701 retained nuclei. These are selective-risk measurements conditional on accepting fewer predictions; the rejected nuclei are not considered solved and coverage falls to 80%. The selective-prediction curve therefore characterizes the accuracy–coverage trade-off rather than performance on a fixed full dataset.

Deep-ensemble uncertainty and selective prediction

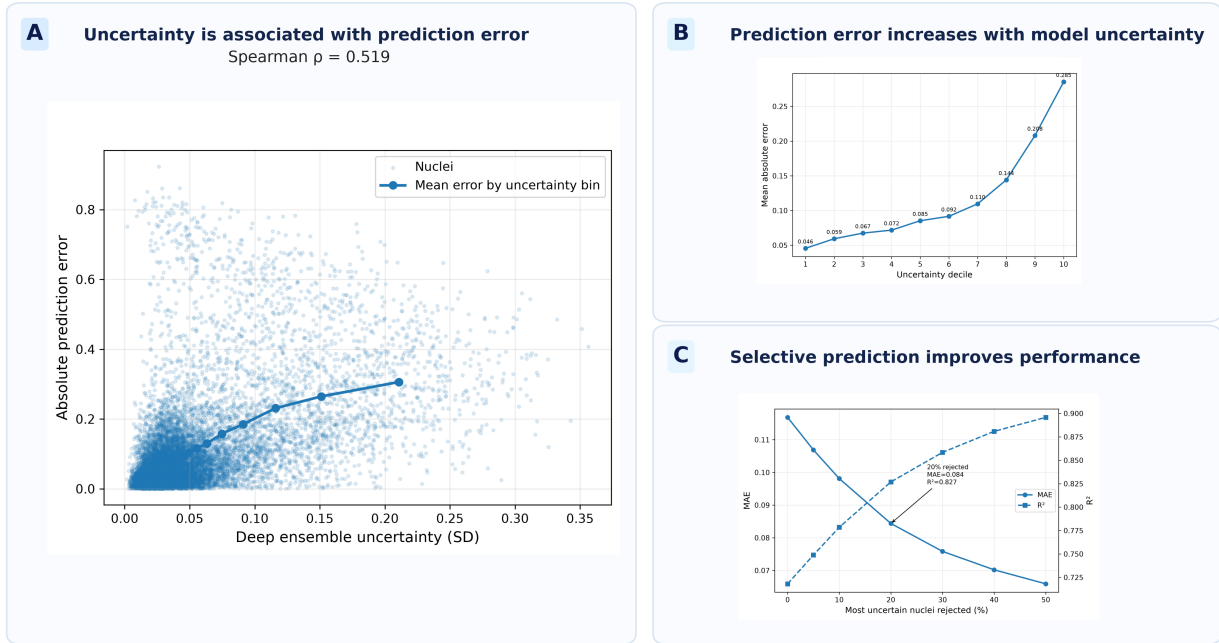


Figure 3: Deep-ensemble uncertainty and selective prediction. (A) Per-nucleus ensemble disagreement versus absolute error; 12,000 randomly sampled nuclei are plotted, but Spearman $\rho = 0.519$ and Pearson $r = 0.493$ use all 30,877 OOF nuclei. (B) Mean absolute error across uncertainty deciles. (C) Selective MAE and R^2 on the retained set as the most uncertain nuclei are rejected. At 20% rejection, MAE is 0.084 and R^2 is 0.827.

7 Longitudinal sensitivity analysis

We separately examine changes in M1–M4 predictions across three acquisition days and assess their association with changing image characteristics. These analyses test temporal sensitivity rather than directly identifying biological differentiation dynamics.

7.1 Protocol and common support

The longitudinal comparison uses 88,458 common observations from six cultures (02A, 05D, 08A, 11D, 14A, and 17D), each represented at Day00, Day01, and Day02. Model-specific longitudinal scores from M1–M4 are analyzed over matching observations and summarized by culture and acquisition day. For M3, the longitudinal score uses the fold-specific development-only latent-PC1 calibration reconstructed from the frozen M3 checkpoints, as detailed in Section 10; the longitudinal observations themselves are not used to fit or orient the latent axis. Contrasts with Day00 are calculated from these model-specific mean scores. The common-support construction prevents differences in the evaluated observation pool from being mistaken for model differences.

No matched longitudinal fluorescence-derived differentiation ground truth is available for these six trajectories. Consequently, the analysis cannot identify which inferred time course best represents the underlying biology.

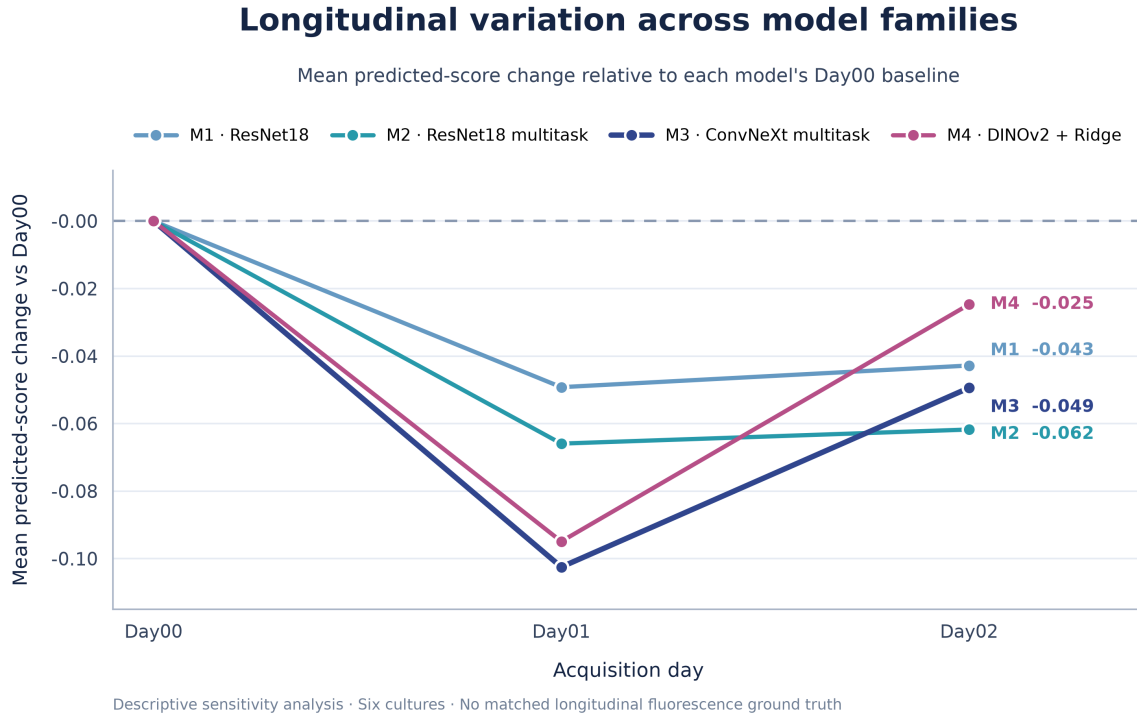
7.2 Model-dependent trajectories

All four model-derived scores exhibit a reduction in their mean value between Day00 and Day01. M1 and M2 remain close to their Day01 level at Day02, whereas M3 and M4 show a larger rebound.

Table 3: Mean longitudinal changes in model-derived differentiation score relative to each model’s Day00 baseline.

Model	Day01–Day00	Day02–Day00
M1	-0.0492	-0.0428
M2	-0.0659	-0.0618
M3	-0.1025	-0.0494
M4	-0.0949	-0.0247

Similar mean curves do not establish agreement at the individual-culture level, where the model-specific temporal responses differ.

**Figure 4: Longitudinal change relative to each model’s Day00 baseline.** Mean model-derived score changes from the common support of six cultures. This is a descriptive sensitivity analysis without matched longitudinal fluorescence ground truth.

7.3 Acquisition confounder sensitivity

For each day transition, Spearman correlations across the six cultures describe the relationship between changes in model predictions and changes in image contrast, normalized Tenengrad focus, and median spots per frame. These variables are acquisition or sampling proxies, not independently randomized interventions. The correlations are descriptive because $n = 6$.

For Day01–Day00, contrast change is weakly associated with M1 and M2 predictions ($\rho = 0.086$ and 0.029) and negatively associated with M3 ($\rho = -0.771$). Changes in spots per frame yield $\rho = -0.543, -0.371, +0.714$, and $+0.600$ for M1–M4, respectively.

For Day02–Day00, normalized Tenengrad change is negatively correlated with M1 ($\rho = -0.943$) and M2 ($\rho = -0.771$), with substantially smaller corresponding associations for M3 ($\rho = -0.086$) and M4 ($\rho = -0.029$). For Day02–Day01, M4 prediction changes correlate with spots per frame ($\rho = +0.829$). The associations vary by model and interval and cannot establish that acquisition changes caused the predicted temporal differences.

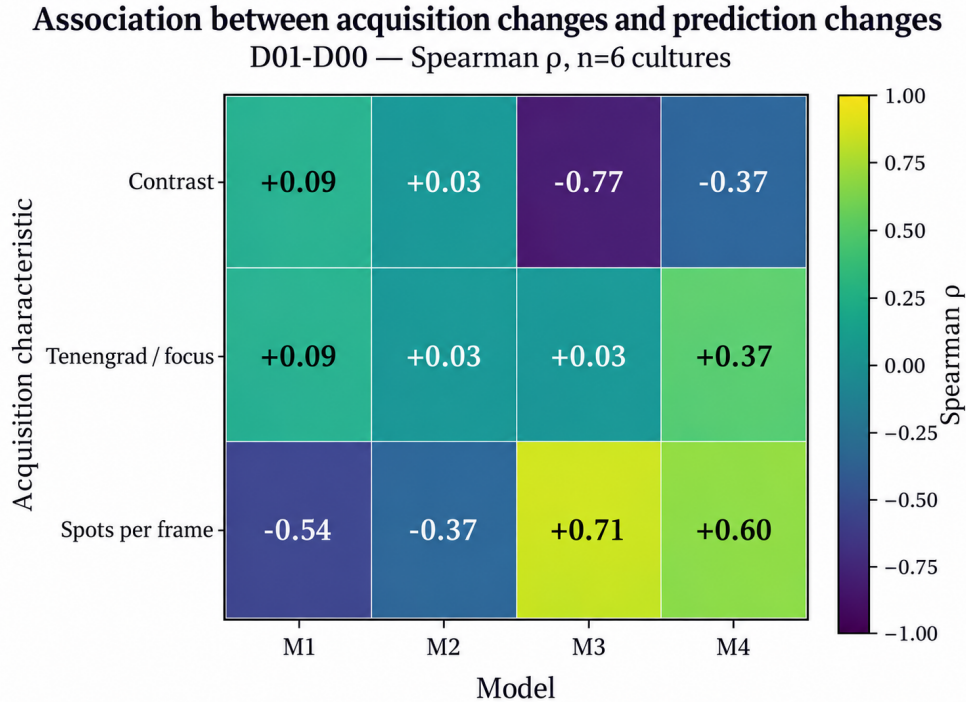


Figure 5: Association between Day01–Day00 prediction changes and acquisition characteristics. Spearman correlations are computed over six culture-level pairs per cell. The coefficients are descriptive, not tests of causality. The full three-interval matrix is provided in Appendix D.

8 Discussion

The static results support three findings. First, learning directly from brightfield crops improves differentiation-label prediction relative to the frozen feature-based M0 baseline. Second, the addition of marker-expression supervision improves ResNet18 while preserving label-free inference. Third, the ConvNeXt-Tiny multitask model gives the largest single-model accuracy gain, while the frozen DINOv2 representation remains below M3 on the primary nucleus-level endpoint.

Deep ensembling adds a modest nucleus-level accuracy gain while retaining strong culture-level performance. Ensemble disagreement also supplies a useful triage signal: uncertainty deciles show increasing observed error, and uncertainty-based abstention reduces error among retained nuclei. This operational benefit does not establish probabilistic calibration, reliable error bounds for individual nuclei, or biological validity outside the evaluated domain.

The longitudinal analysis draws an important boundary around the static result. Although predicted mean scores vary systematically over days, the trajectories are representation-dependent and their associations with image characteristics vary between transitions. The absence of matched longitudinal marker truth precludes choosing a biologically correct model trajectory on these data. Static accuracy should therefore not be presented as a validated longitudinal biological readout.

9 Limitations

These limitations delimit what the reported experiments do and do not establish.

1. **No independent external validation set.** Held-out-culture OOF evaluation does not establish transfer to new microscopes, laboratories, acquisition settings, or hiPSC cell lines.
2. **Different published benchmark support and split.** The historical study used 26,213 nuclei

from 49 cultures and nucleus-level random splitting; the local comparison uses 30,877 nuclei from 48 cultures and held-out cultures. Their reported metrics are not directly matched estimates.

3. **No exhaustive hyperparameter optimization.** The reported training configurations are fixed development configurations rather than the result of a systematic search over learning rates, loss weights, augmentation strengths, or architecture-specific hyperparameters.
4. **Upstream localization and segmentation are outside the primary endpoint.** The evaluated regressors operate on per-nucleus crops; the report does not quantify full-volume detection and crop-extraction error on a new instrument.
5. **No matched longitudinal fluorescence ground truth.** Model-predicted time courses cannot be validated as actual differentiation dynamics for the six longitudinal cultures.
6. **Small longitudinal culture count.** Confounder correlations use only six culture-level pairs and are descriptive; multiple variables and transitions are examined without causal identification.
7. **Potential acquisition sensitivity.** Observed associations with contrast, focus, and density proxies may reflect imaging or biological differences and do not isolate their contributions.
8. **Deep-ensemble cost.** Five network evaluations are needed for each TTA view to obtain an ensemble output, increasing training, storage, and inference requirements compared with a single M3 network.
9. **No formal uncertainty calibration.** Ensemble disagreement is associated with observed error and supports selective prediction, but calibrated probabilities, coverage guarantees, or individual-error bounds have not been established.

10 Reproducibility

All final experiments use frozen five-fold culture assignments. For the M3 deep ensemble, five training seeds (42–46) are used for each outer fold, while the internal train/validation split seed is fixed at 42. Consequently, the data partitions remain identical across ensemble members and only the training stochasticity changes, yielding 25 trained models in total.

The original experiments were executed in a Python 3.11 `oocnet` Conda environment. The public reproducibility release provides a clean `bright2nuc` environment specification based on Python 3.11 through `environment.yml`, together with pinned Python dependencies in `requirements.txt`. The recorded M3 run used PyTorch 2.14.0+cu126, torchvision 0.29.0+cu126, timm 1.0.30, and an NVIDIA GeForce RTX 4080 Laptop GPU. The clean release environment was separately validated with CUDA available through PyTorch.

The final M3 experiments use a fixed training configuration: a maximum of 30 epochs, early-stopping patience of five epochs, microbatch size 32, gradient accumulation over eight steps, AdamW optimization, encoder and prediction-head learning rates of 10^{-4} and 3×10^{-4} , respectively, and weight decay 0.05. These values describe the executed configuration and are not presented as globally optimized hyperparameters.

The reproducibility release contains the target-construction and preprocessing scripts, frozen culture-fold definitions, per-model training code, inference and aggregation code, deep-ensemble uncertainty analysis, longitudinal sensitivity analysis, and figure-generation scripts. Compact frozen numerical results and publication-ready figures are included directly in the repository.

Large generated artefacts are intentionally excluded from version control. These include neural-network checkpoints, cached image arrays, extracted embeddings, fitted Ridge probes, latent-PC1 calibrators, and intermediate prediction tables. The corresponding artefacts can be regenerated from the frozen scripts and documented data layout. In particular, the M3 longitudinal analysis

uses latent-PC1 calibrators reconstructed from the frozen M3 checkpoints. Out-of-fold embeddings are first extracted, after which fold-specific PCA-based calibrators are fitted using development data only. The longitudinal data are not used to fit or orient these latent axes. This is why the longitudinal M3 quantity is described as a model-derived score rather than as an independent longitudinal fluorescence measurement.

For image-only inference with M1–M3, the input contract is a per-nucleus $16 \times 64 \times 64$ brightfield crop. Brightfield normalization parameters are estimated from the corresponding training subset and stored with the trained model configuration. For M2 and M3, marker-normalization parameters are training-only quantities used for auxiliary supervision and are not inference inputs. The M3 deep-ensemble output comprises the mean differentiation-score prediction and the sample standard deviation across ensemble members.

The reported out-of-fold benchmark can be reconstructed by rerunning the frozen training and evaluation pipeline with the released fold definitions, preprocessing code, model configurations, and random seeds. Because trained checkpoints are generated artefacts rather than version-controlled files, reproducing the exact archived numerical outputs from previously trained weights requires the corresponding original checkpoints; retraining from source reconstructs the experimental procedure but bitwise-identical model parameters are not guaranteed across hardware, CUDA, or library implementations.

A prospective single-system deployment would require a separately defined training and validation procedure on the full development data and is not evaluated in this report.

11 Data, licensing, and ethics

Bright2Nuc images are publicly accessible through Zenodo and should be attributed to the original study and dataset [1]. Public availability is not equivalent to unrestricted redistribution. Before distributing the code, original images, derived crop archives, or weights, their respective licenses and attribution conditions must be confirmed. The released repository should provide dataset links and an acquisition script or manifest when direct redistribution is not permitted.

This project performs secondary computational analysis of existing hiPSC imaging experiments. It recruits no new human participants, collects no new biological samples, and does not use clinical patient records. Its outputs are research predictions of a marker-derived label, not clinical diagnoses or validated biological time-course measurements.

12 Conclusion

We present a brightfield-only per-nucleus differentiation-score predictor evaluated on held-out cultures. The multitask ConvNeXt-Tiny reaches nucleus-level $R^2 = 0.693$ and $\text{MAE} = 0.121$. A five-member-per-fold ensemble improves these values to $R^2 = 0.718$ and $\text{MAE} = 0.117$ while providing ensemble-disagreement uncertainty. Rejecting the 20% most uncertain predictions yields $R^2 = 0.827$ and $\text{MAE} = 0.084$ on the retained 80% of nuclei.

These results demonstrate predictive information in brightfield morphology for the marker-derived differentiation target on the evaluated Bright2Nuc cultures. The next validation priorities are an independent biological test set, sensitivity to new acquisition settings, formal uncertainty assessment, and matched longitudinal fluorescence measurements. Until those studies are completed, longitudinal predictions are best viewed as an exploratory sensitivity analysis.

A Fold-level static results

Table 4: Mean and standard deviation of fold-level nucleus metrics. These are means of five fold-specific values, not the global pooled OOF statistics in Table 1.

Model	Mean R^2	SD R^2	Mean MAE	SD MAE
M0	0.225	0.246	0.235	0.028
M1	0.398	0.093	0.220	0.026
M2	0.437	0.157	0.208	0.021
M3	0.682	0.133	0.122	0.027
M4	0.438	0.128	0.211	0.023

B Individual deep-ensemble members

Table 5: Global OOF nucleus-level performance of the five M3 members and their mean prediction.

Prediction	R^2	MAE	Pearson	Spearman
Seed 42	0.693	0.121	0.836	0.765
Seed 43	0.660	0.126	0.822	0.760
Seed 44	0.680	0.121	0.830	0.759
Seed 45	0.659	0.127	0.815	0.754
Seed 46	0.702	0.119	0.842	0.767
Ensemble mean	0.718	0.117	0.848	0.777

The ensemble is retained rather than selecting the best seed after inspecting the held-out results; retrospective single-seed selection would introduce an evaluation bias.

C Uncertainty diagnostics

Table 6: Mean absolute error by increasing decile of ensemble standard deviation.

Decile	1	2	3	4	5	6	7	8	9	10
MAE	.046	.059	.067	.072	.085	.092	.110	.144	.208	.285

D Longitudinal confounder correlations

Table 7: Spearman correlations between acquisition-characteristic changes and prediction changes. Values are descriptive ($n = 6$ cultures).

Transition	Characteristic	M1	M2	M3	M4
Day01–Day00	Contrast	0.086	0.029	-0.771	-0.371
	Tenengrad/focus	0.086	0.029	0.029	0.371
	Spots/frame	-0.543	-0.371	0.714	0.600
Day02–Day01	Contrast	-0.486	-0.543	-0.143	0.543
	Tenengrad/focus	-0.257	0.257	-0.657	0.371
	Spots/frame	-0.486	0.029	-0.314	0.829
Day02–Day00	Contrast	0.314	0.257	-0.314	0.714
	Tenengrad/focus	-0.943	-0.771	-0.086	-0.029
	Spots/frame	0.143	0.771	-0.143	0.486

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